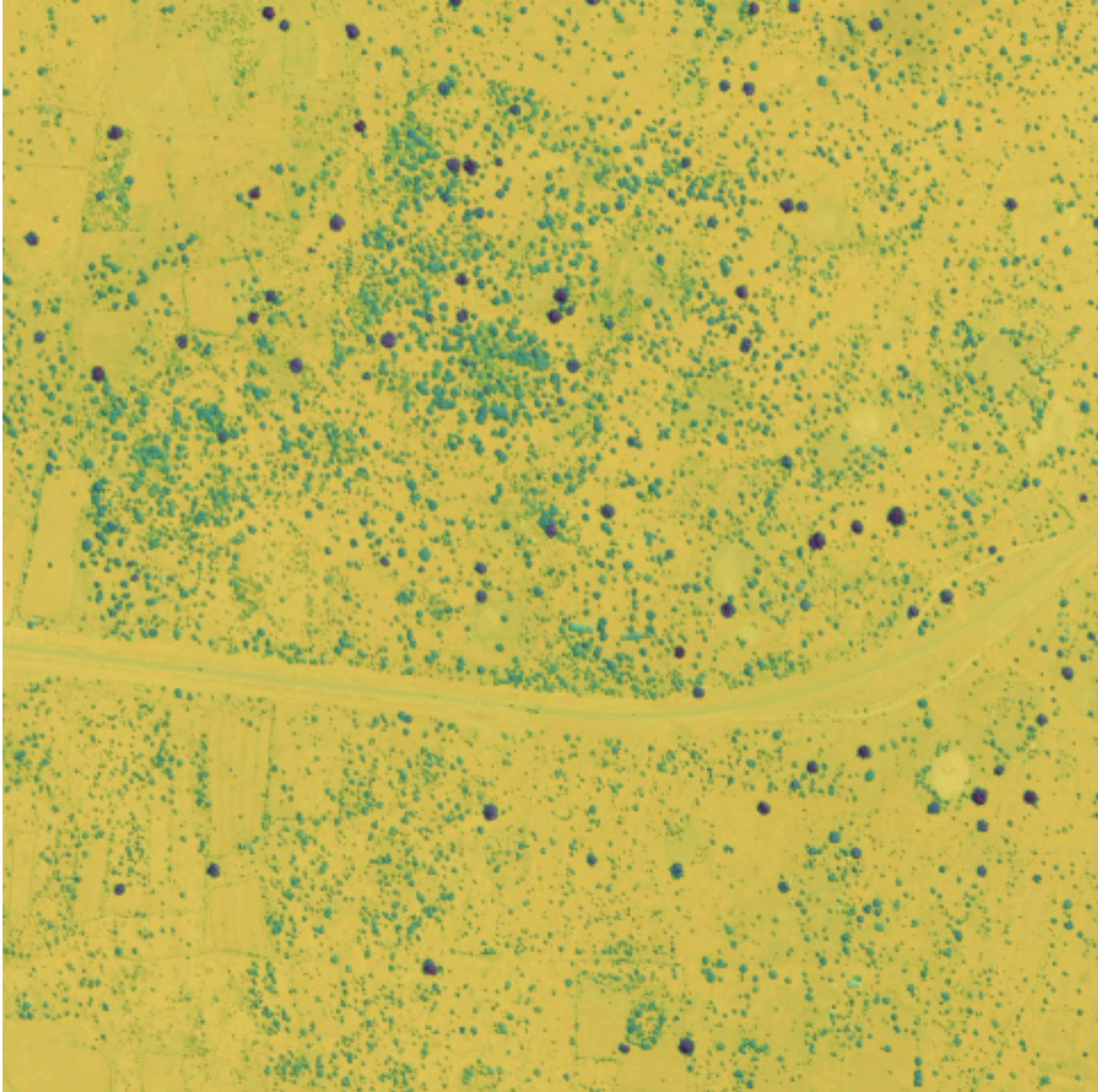


Practical 1

ALS 1: Introduction to Airborne Laser Scanning



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1. Introduction

Airborne laser scanning (ALS), also known as LiDAR (Light Detection and Ranging), is a remote sensing technique used to collect highly accurate and detailed information about the Earth's surface and its features. It involves the use of laser pulses emitted from an aircraft-mounted sensor to measure the distance between the sensor and the objects on the ground.

The basic principle of ALS involves sending out laser pulses and measuring the time it takes for the pulse to return after hitting an object. By calculating the travel time and knowing the speed of light, the distance between the sensor and the object can be determined with great precision. This process generate a dense point cloud of 3D coordinates (Z,Y,Z) representing the surface of the Earth.

ALS provides several advantages over traditional surveying methods. It allows for rapid data collection over large areas and terrain that may be inaccessible or difficult to reach on foot. It enables the acquisition of high-resolution elevation data, capturing detailed information about the topography of the landscape, such as terrain slopes, vegetation height, and the location of buildings, power lines, and other objects.

The collected ALS data can be processed and analysed to create raster digital elevation models (DEMs), which are widely used for various applications, such as urban planning, forestry, environmental monitoring, hydrological modelling, and natural resource management.

In the first part of this exercise, we will study and visualise typical ALS data products, such as Digital Terrain Model, Digital Surface Model and Canopy Height Model in QGIS. In the second part of the practical, we will study basic properties of ALS data and go through typical processing steps to process ALS data into those products. The second part is made in R software environment, and it also includes a short introduction to R programming language for those who use it for the first time.

2. Visualizing ALS data products in QGIS

2.1. Digital Terrain Model (DTM)

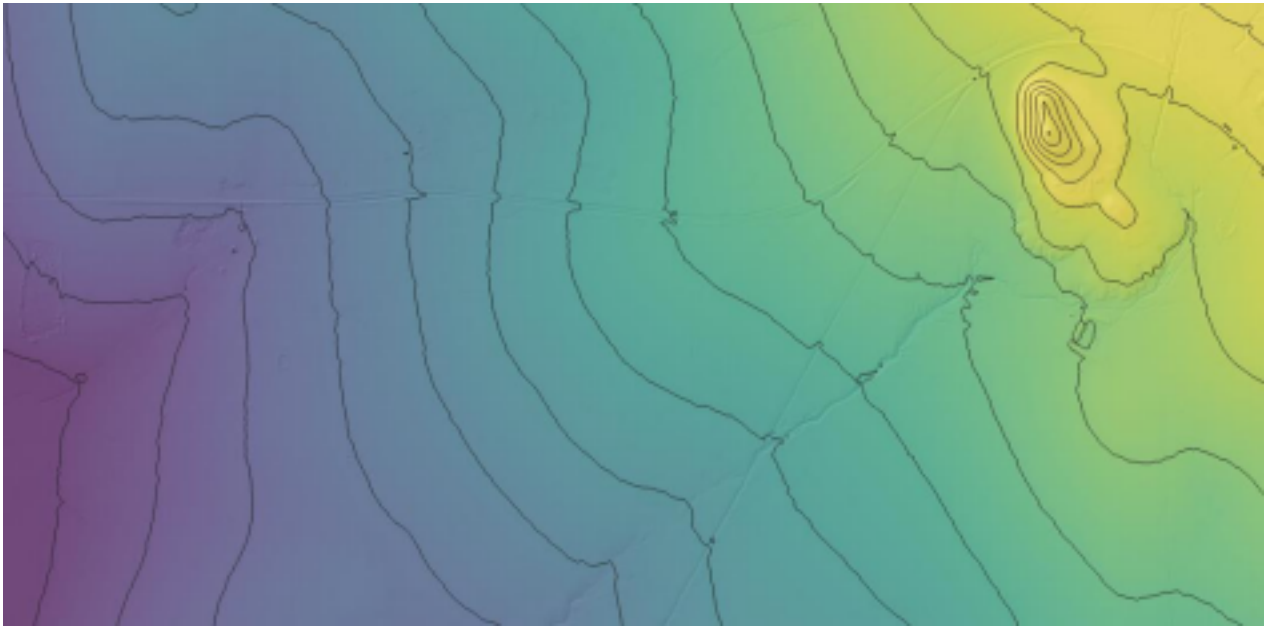
Digital Terrain Model (DTM) is the most basic data product derived from ALS data. It is based on the lowest laser points that represent Earth's bare surface. It is typically represented as a raster (grid) of elevation values, where each cell contains a numerical value representing the elevation at that location. DTMs are commonly used in various application fields such as geography, geology, cartography, urban planning, hydrology, and environmental modelling to analyse and visualize terrain features, assess slope and aspect, and support spatial analyses.

Open the DTM of the Taveta study area ([Taveta_DTM.tiff](#)) as a new raster file in QGIS ([Layer > Add Layer > Add Raster Layer...](#)). See Layer properties (right click raster name and select [Properties](#)).

Under [Symbology](#), you can change the rendering type. [Singleband pseudocolor](#) is usually providing the best color scale. Under [Min / Max Value Settings](#), you can find several options to set Min and Max values automatically. You may also set those manually. Try also [Hillshade](#) and [Contours](#) options.

We can also combine all three rendering types. Make two copies of the layer (right click raster name and select [Duplicate Layer](#)). Use [Singleband pseudocolor](#) for the first layer. Set [Global Opacity](#) in [Transparency](#) menu under Layer properties to 60%. Use [Hillshade](#) for the second layer, and use [Contour](#) for the third layer ([Contour interval](#) e.g. 10 m).

Finally, order layers so that contours are on the top, Pseudocolor image in the middle and Hillshade image in the bottom. It should look like image below.



Change parameters so that you are happy with the visualization. You can also group these three layers so that you view and hide all three layers at once (select layers using **Ctrl** and left button of the mouse > right click and **Group Selected**).

2.2. Digital Surface Model (DSM)

A Digital Surface Model (DSM) represents the highest points in the landscape. DSM thus includes three-dimensional information of the ground and objects on the ground, such as trees and buildings.

Open the DSM of the Taveta study area ([Taveta_DSM.tif](#)). Try Pseudocolor and Hillshade options for rendering. Observe how DSM differs from DTM.

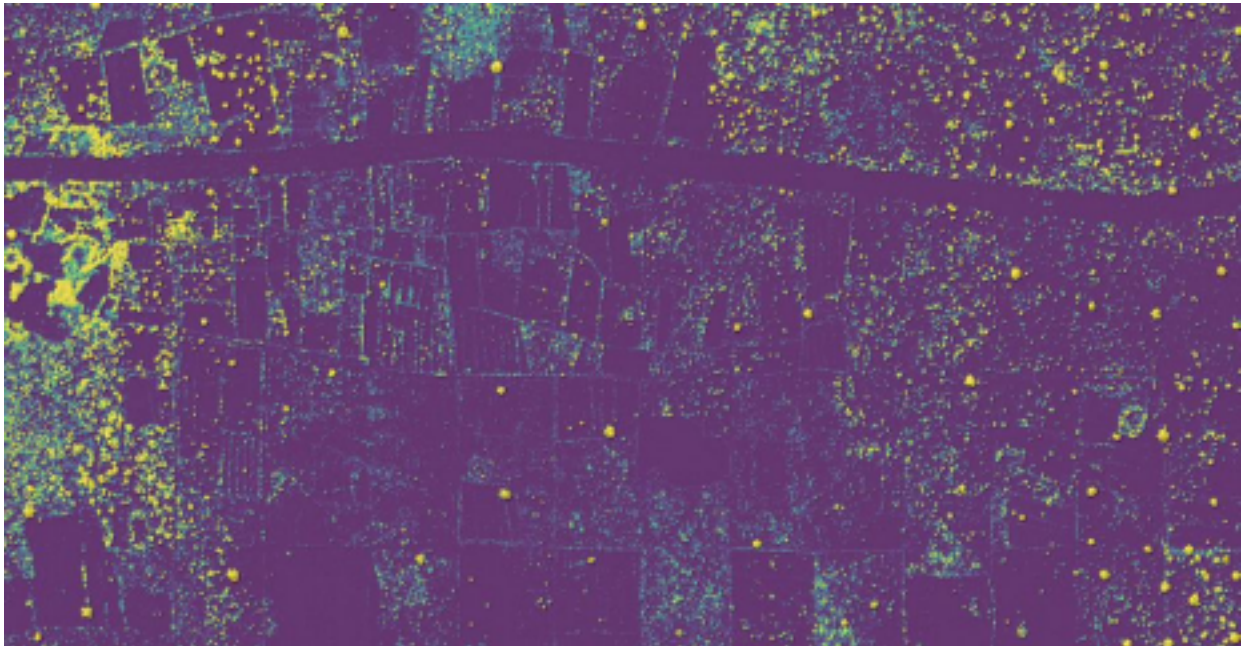
2.3. Calculate CHM

A Canopy Height Model (CHM) represents the height of the objects on the ground. The height of the canopy can be calculated by subtracting the DTM (bare Earth elevations) from DSM (surface defining the highest points). The raster layers must have the same spatial resolution.

To create a CHM, use the Raster Calculator (**Raster > Raster Calculator**) to subtract the DTM from the DSM. Name your output layer as [Taveta_CHM.tif](#).

Visualise the layer using Pseudocolor rendering. Observe minimum and maximum values of the resulting layer. Study layer values also using Identify Features tool, which tell you raster cell values for the selected layer. What do high values correspond to?

CHM can be visualized effectively similarly to DTM by combining Pseudocolor and Hillshade rendering (example below) but both of them are quite effective also separately.



In the following part, you will learn what point clouds are like, and what steps are needed to generate abovementioned raster products.